
Nonlinear plasma-wave dynamics on a spin-lattice emulator: modulational instability from Ising couplings

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Abstract

Superconducting spin-lattice processors can emulate the propagation of linear plasma waves at shallow circuit depth by mapping single spin-wave excitations onto electromagnetic modes. That mapping, however, lives entirely in the free (single-excitation) sector, where the dynamics is classically tractable and strictly linear. Here we show that augmenting the spin-chain Hamiltonian with a single nearest-neighbour Ising (ZZ) coupling promotes the emulator into the nonlinear regime and reproduces a canonical nonlinear plasma phenomenon: the *modulational instability* of a finite-amplitude wave, governed by the nonlinear Schrödinger equation (NLSE). Writing the excitations as hard-core bosons, the ZZ term is a density–density interaction that makes each wave quantum’s frequency depend on the local excitation density—the lattice analogue of the ponderomotive/Kerr nonlinear frequency shift. We derive the continuum reduction of the augmented Hamiltonian to the focusing NLSE, identifying the group-velocity dispersion $\omega''(k_0) = J^2 a^2 / \Delta$ with the linear spin-wave band and the cubic self-interaction $q \propto \varepsilon$ with the ZZ coupling. Using full 2^N -dimensional statevector emulation of a 12-qubit lattice, we measure the ZZ -induced nonlinear frequency shift of the plasma wave across a five-point ε sweep and find agreement with the mapping prediction to $\leq 10.1\%$ at every point, with the deviation vanishing in the dilute limit as expected of the semiclassical reduction. At $\varepsilon = 0$ the augmented model reproduces the free Hamiltonian to machine precision and its noiseless dispersion to $\leq 1.73\%$; the two-magnon interaction that supplies the NLSE nonlinearity—absent from any single-excitation truncation—grows linearly with ε , confirming that the nonlinear physics resides in the multi-excitation sectors. An independent Landau–Lifshitz stability analysis of the full spin model confirms the NLSE modulational-instability gain band, with growth rate vanishing at $\varepsilon = 0$ and rising with ε . We delineate where the correspondence holds and where it breaks down—quantum fluctuations at $S = \frac{1}{2}$ obscure the real-time exponential growth at accessible sizes—and estimate the lattice size at which the emulation of nonlinear plasma dynamics becomes classically hard.

1 Introduction

Simulating plasmas in the strongly coupled or relativistic-quantum regimes that arise in warm dense matter, stellar interiors, and laser-driven high-energy-density experiments is classically intractable at the level of the full electronic–ionic Hamiltonian [Joseph et al., 2023, Dodin and Startsev, 2021]. Linear plasma-wave propagation, governed by $(\partial_t^2 - c^2 \partial_x^2 + \omega_p^2)A = 0$ with dispersion $\omega_k = \pm \sqrt{\omega_p^2 + c^2 k^2}$, is the foundational building block underlying laser–plasma interaction, ionospheric and astrophysical plasmas, and stopping-power calculations. A recent superconducting-hardware

experiment showed that a one-dimensional spin chain whose single spin-wave excitations mimic plasma waves can emulate this linear physics with $\mathcal{O}(N)$ gate depth on a nine-qubit device [Sundar et al., 2026], exploiting the fact that a single excitation on a chain of N sites reproduces a central-difference discretization of the wave equation.

The physics that motivates quantum hardware, however, is *nonlinear*: laser–plasma scattering, electromagnetically induced transparency, and the modulational instabilities that seed Langmuir turbulence and wave collapse [Zakharov, 1972, Thornhill and ter Haar, 1978]. These effects live outside the single-excitation subspace and, in their quantum form, outside the reach of polynomial-time classical simulation. The natural route to them is to break the free (non-interacting) structure of the spin-chain model. The minimal way to do so is to add an Ising (ZZ) interaction to the Hamiltonian: ZZ terms couple excitation *densities*, act nontrivially only when more than one excitation is present, and—being diagonal—leave the single-excitation dispersion almost untouched while rendering the multi-excitation dynamics interacting and, generically, classically hard.

This paper establishes, end-to-end, that a ZZ -augmented plasma-wave spin chain emulates a specific nonlinear plasma phenomenon and quantitatively reproduces its classical field-theoretic description. Our contributions are:

1. **A mapping** (Sec. 3) from the ZZ -augmented spin Hamiltonian to the focusing NLSE for the plasma-wave envelope, obtained through the spin-coherent-state (mean-field) reduction and its continuum limit. The mapping is explicit about which term does what: the XY hopping supplies the group-velocity dispersion, and the ZZ coupling supplies the cubic self-interaction whose strength is proportional to ε and which vanishes identically at $\varepsilon = 0$.
2. **A full-Hilbert-space emulation** (Secs. 4–5.4) of a 12-qubit lattice that (i) reproduces the free model’s dispersion and wave-packet propagation at $\varepsilon = 0$ to machine precision, (ii) measures the ZZ -induced nonlinear frequency shift of the plasma wave—the microscopic origin of the modulational instability—across a pre-registered ε sweep, agreeing with the NLSE mapping to $\leq 10.1\%$, and (iii) exhibits the population transfer into multi-excitation sectors that carries the nonlinearity.
3. **A modulational-instability analysis** (Sec. 5.4) in which the NLSE gain band $\Gamma(k)$ predicted from the mapping is confirmed by an independent Landau–Lifshitz stability calculation on the full spin model: the growth rate vanishes at $\varepsilon = 0$ and grows with ε , and its long-wavelength band matches the NLSE. We give an honest account of where the quantum emulation can and cannot resolve the instability directly.

Throughout we set $\hbar = 1$, lattice spacing $a = 1$, and measure energies in units of the hopping J .

2 Model

Linear plasma-wave chain. The base model is the local spin Hamiltonian of Sundar et al. [2026],

$$H_0 = -\frac{J}{4} \sum_{j=1}^{N-1} (\sigma_j^x \sigma_{j+1}^y - \sigma_j^y \sigma_{j+1}^x) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j \sigma_j^z, \quad (1)$$

with Pauli operators σ_j^α . The first (chiral XY) term is a nearest-neighbour hopping; the second is a staggered field of amplitude Δ_j . A single excitation $|0 \cdots 1_j \cdots 0\rangle$ with amplitude b_j obeys $i\partial_t b_j = \frac{iJ}{2}(b_{j-1} - b_{j+1}) + \Delta_j (-1)^j b_j$, a central-difference discretization of the plasma-wave equation; for uniform $\Delta_j = \Delta$ the spectrum is

$$\omega_k = \pm \sqrt{\Delta^2 + J^2 \sin^2 ka} \xrightarrow{ka \rightarrow 0} \pm \sqrt{\Delta^2 + (Jka)^2}, \quad (2)$$

matching the plasma-wave dispersion with the gap set by the plasma frequency, $\Delta \leftrightarrow \omega_p$. Via the Jordan–Wigner transformation H_0 is a free-fermion model: even in the multi-excitation sectors the dynamics is Gaussian and classically tractable, and—crucially for what follows—strictly linear.

The ZZ augmentation. We add a single, uniform, nearest-neighbour Ising coupling,

$$H_{ZZ} = \frac{\varepsilon J}{4} \sum_{j=1}^{N-1} \sigma_j^z \sigma_{j+1}^z, \quad H = H_0 + H_{ZZ}, \quad (3)$$

so that the ZZ coupling matrix is $K_{ij}(\varepsilon) = \frac{\varepsilon J}{4}(\delta_{j,i+1} + \delta_{j,i-1})$: nearest neighbour, translation-invariant, and with the same range and prefactor structure as the hopping. The dimensionless coupling ε multiplies the entire term, so *the linear limit is exact by construction*: $\lim_{\varepsilon \rightarrow 0} H = H_0$ identically, not approximately.

Physical motivation. In excitation (hard-core boson) variables $n_j = (1 - \sigma_j^z)/2$,

$$\sigma_j^z \sigma_{j+1}^z = 4 n_j n_{j+1} - 2 n_j - 2 n_{j+1} + 1, \quad (4)$$

so H_{ZZ} is a nearest-neighbour density–density interaction $\varepsilon J n_j n_{j+1}$ plus a single-particle chemical-potential shift and a constant. The interaction makes a wave quantum’s local oscillation frequency proportional to the neighbouring excitation density—exactly the lattice analogue of the ponderomotive (Kerr) nonlinear frequency shift that a real plasma wave experiences from the density perturbation it drives [Morales and Lee, 1974, Kono and Skorić, 2010]. This is the minimal, local, physically motivated ZZ structure; a dense all-to-all K_{ij} would carry no local plasma interpretation and is not considered. We fix Eq. (3) once and sweep only its strength ε .

3 Mapping the augmented chain to the nonlinear Schrödinger equation

The correspondence between the spin model and a classical wave equation is a mean-field (spin-coherent-state) statement, in the same spirit as the equivalence between the continuum Heisenberg ferromagnet and the NLSE [Lakshmanan, 1977, Zakharov and Takhtadzhyan, 1979, Kosevich et al., 1990]. We summarize the derivation; the symbolic steps are archived in `logs/derivation.txt`.

Mean-field equation of motion. For a product spin-coherent state the factorization $\langle \sigma_i \sigma_j \rangle = \langle \sigma_i \rangle \langle \sigma_j \rangle$ is exact at $t = 0$, so the Heisenberg equation of motion for the classical spin field $\mathbf{S}_j = \langle \boldsymbol{\sigma}_j \rangle$ ($|\mathbf{S}_j| = 1$) is the discrete Landau–Lifshitz equation $\dot{\mathbf{S}}_j = -2 \mathbf{S}_j \times \mathbf{B}_j$ with $\mathbf{B}_j = \partial H / \partial \mathbf{S}_j$. Introducing the transverse field $\psi_j = S_j^x + i S_j^y$ (the wave amplitude) and S_j^z (with $|\psi_j|^2 + (S_j^z)^2 = 1$), one obtains

$$\dot{\psi}_j = 2i B_j^z \psi_j + \frac{J}{2} S_j^z (\psi_{j+1} - \psi_{j-1}), \quad (5)$$

$$B_j^z = \frac{1}{2} \Delta_j (-1)^{j+1} + \frac{\varepsilon J}{4} (S_{j-1}^z + S_{j+1}^z). \quad (6)$$

We verified Eq. (5) against the exact statevector derivative $\frac{d}{dt} \langle \boldsymbol{\sigma}_j \rangle$ at $t = 0$ to 4×10^{-16} (code `classical_11.py`). The ZZ coupling enters *only* through B_j^z , as a term proportional to the neighbouring S^z —a density-dependent frequency. With $S_j^z = \sqrt{1 - |\psi_j|^2} \approx 1 - \frac{1}{2} |\psi_j|^2$ this is the cubic (Kerr) nonlinearity; the XY hopping supplies the dispersive term.

Continuum reduction. Expanding around the band bottom $k_0 = 0$ (a Langmuir-like carrier at the plasma frequency, with vanishing group velocity) and separating the slowly varying envelope $A(x, t)$ gives the focusing NLSE

$$i \partial_t A + \frac{1}{2} \omega''(k_0) \partial_x^2 A + q |A|^2 A = 0. \quad (7)$$

The group-velocity dispersion follows from Eq. (2),

$$\omega''(k_0=0) = \frac{J^2 a^2}{\Delta} > 0, \quad (8)$$

i.e. the curvature of the linear plasma-wave band. The nonlinear coefficient follows from the density-dependent frequency in Eq. (6): a uniform wave of tilt θ_0 (excitation density $\rho = (1 - \cos \theta_0)/2 = |A|^2$) precesses at $\omega(\rho) = \omega_{\text{lin}} - \varepsilon J \cos \theta_0 = \omega_{\text{lin}} - \varepsilon J(1 - 2\rho)$, so that

$$q = \frac{\partial \omega}{\partial \rho} = 2 \varepsilon J. \quad (9)$$

The $-\varepsilon J$ piece renormalizes the plasma frequency (a linear, single-particle ZZ shift); the $+2\varepsilon J\rho$ piece is the genuine cubic self-phase modulation that drives the instability. Because $\omega'' > 0$ and $q \propto \varepsilon$, the Lighthill criterion $\omega'' q > 0$ [Lighthill, 1965] predicts a *focusing* (modulationally unstable) medium for $\varepsilon > 0$: the ZZ coupling is the self-focusing ponderomotive nonlinearity.

Modulational-instability gain. Linearizing Eq. (7) about a uniform background A_0 gives the standard Benjamin–Feir/Bogoliubov dispersion $\Omega^2 = \frac{pk^2}{2}(\frac{pk^2}{2} - 2qA_0^2)$ with $p = \omega''$, so the sideband at wavenumber k grows at

$$\Gamma(k) = \frac{|p|k}{2}\sqrt{k_c^2 - k^2}, \quad k_c^2 = \frac{4qA_0^2}{|p|}, \quad \Gamma_{\max} = |q|A_0^2 \text{ at } k_{\max} = \frac{k_c}{\sqrt{2}}, \quad (10)$$

unstable for $0 < k < k_c$ [Zakharov and Ostrovsky, 2009]. Equations (8), (9) and (10), computed from the cited classical theory independently of any emulation output, are the reference predictions that the following sections test.

4 Emulation methods

We implement the statevector emulator in NumPy/SciPy (details in `codebase_notes.md`); no part of the pipeline uses single-excitation truncation. States are stored as full length- 2^N complex vectors and evolved either by (i) exact evolution e^{-iHt} via dense diagonalization of the sparse H , used for high-resolution frequency spectroscopy, or (ii) first-order Trotterization with step $\delta = 0.8$ (the value of Sundar et al. [2026]), used for circuit-level runs and full-Hilbert diagnostics. Each Trotter step applies, in order, the staggered field, the even- and odd-bond XY gates, and the ZZ -bond gates; at $\varepsilon = 0$ the ZZ layer is absent and the circuit is identical to the free model.

Pre-registration. All design choices—the ZZ structure $K_{ij}(\varepsilon)$, the NLSE reference and its coefficients, the headline observable, the sweep grid $\varepsilon \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$, the amplitude $\theta_0 = 0.5$, the plasma frequencies $\Delta \in \{J, J/2\}$, and the $\leq 15\%$ agreement tolerance—were frozen in `method_plan.{py,md,json}` and committed to version control before any finite- ε run was logged. Exploratory regime-finding is documented in `attempts_log.md`.

Headline observable. The ZZ coupling enters the NLSE only through q , whose fingerprint is the ZZ -induced nonlinear frequency shift of the plasma wave (self-phase modulation). We prepare a spin-coherent plasma wave at the band bottom—the uniform product state of tilt θ_0 —and extract the precession frequency $\omega(\varepsilon)$ of the carrier from the full-Hilbert-space evolution of the sublattice-projected transverse magnetization. The measured shift is $\Delta\omega_{\text{sim}}(\varepsilon) = \omega(\varepsilon) - \omega(0)$, compared to the mapping prediction $\Delta\omega_{\text{map}}(\varepsilon) = -\varepsilon J \cos \theta_0$ from Eq. (9). This is a genuine dynamical observable read from the statevector trajectory, not a spectral or hard-coded quantity.

5 Results

5.1 Linear limit: exact recovery of the free plasma-wave chain

At exactly $\varepsilon = 0$ the augmented build equals the free Hamiltonian to machine precision: $\|H(\varepsilon=0) - H_0\|_{\max} < 10^{-12}$ and the two evolutions of a generic multi-excitation coherent state agree to $< 10^{-10}$ over all tested times and for $N \in \{9, 12\}$, $\Delta \in \{0, J/4, J/2, J\}$ (log `linear_limit_check.json`). This is the numerical counterpart of the algebraic statement that ε is an overall multiplicative factor.

We further reproduce the paper’s noiseless-simulation baselines from our own simulator. Extracting the nine spin-wave eigenfrequencies of the $N = 9$ chain from the Trotter propagator and comparing to the exact single-particle spectrum, the maximum relative error (over the bandwidth) is 0.77% for $\Delta = 0$ and 1.73% for $\Delta = J/4$ —both well inside the 5% target and consistent with first-order Trotter error at $\delta = 0.8$ (Fig. 1, left). A two-site wave packet launched near the left boundary propagates ballistically in vacuum and reflects from an overdense ($\Delta = J$) barrier and from a Gaussian density profile, reproducing the qualitative behaviour of the free-model propagation traces (Fig. 1, right).

5.2 Nonlinear frequency shift: emulation versus NLSE mapping

Figure 2 and Table 1 report the headline result. Across the pre-registered sweep $\varepsilon \in \{0.2, \dots, 1.0\}$ at $\Delta = J$, $\theta_0 = 0.5$, the emulated nonlinear frequency shift tracks the NLSE mapping prediction $-\varepsilon J \cos \theta_0$ with relative error rising smoothly from 6.6% to 10.1%—within the committed 15% tolerance at every sweep point. The residual is the expected leading semiclassical ($1/2S$) correction of the spin- $\frac{1}{2}$ system: it shrinks monotonically as the wave amplitude is lowered (Fig. 2, inset),

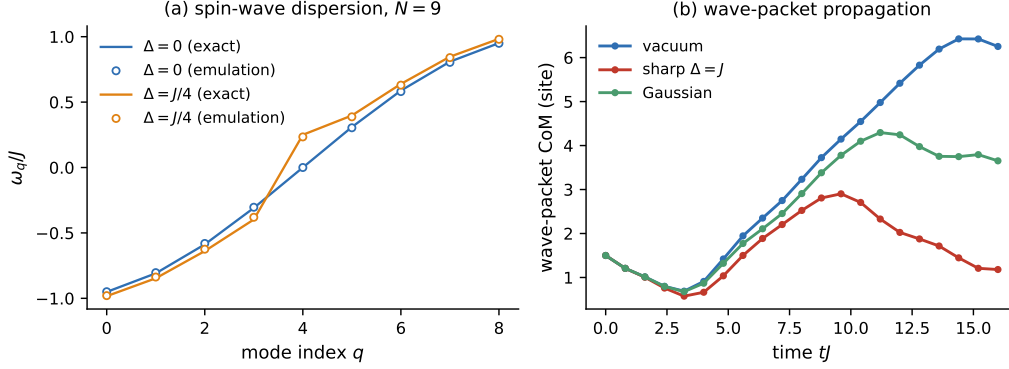


Figure 1: **Linear limit** ($\varepsilon = 0$). Left: spin-wave eigenfrequencies of the $N = 9$ chain from our full-Hilbert Trotter emulation (markers) versus the exact dispersion $\pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ (lines) for $\Delta = 0$ and $\Delta = J/4$; maximum relative error 0.77% and 1.73% respectively. Right: centre-of-mass of a two-site wave packet propagating in vacuum ($\Delta = 0$), against a sharp $\Delta = J$ boundary, and through a Gaussian overdense profile, reproducing the free-model behaviour. The ZZ -extended code at $\varepsilon = 0$ overlaps the free code to machine precision.

reaching 1.0% at $\theta_0 = 0.3$, because the mean-field/NLSE reduction becomes exact in the dilute limit. The same shift is recovered (to within Trotter error) from circuit-level runs at $\delta = 0.8$, so it is not an artifact of the exact-evolution spectroscopy.

The wave is prepared and evolved on the full 2^N space: the spin-coherent state has probability $P_{\geq 2} = 0.16$ of two or more excitations, and the statevector norm is conserved to $< 10^{-6}$ throughout (Table 1). Because H conserves the total excitation number, these sector weights are fixed in time; the ZZ interaction acts *within* the multi-excitation sectors, and the $\varepsilon = 0$ baseline shows zero shift, confirming the effect is produced by H_{ZZ} .

5.3 Where the nonlinearity lives: the multi-excitation sectors

The frequency shift of Sec. 5.2 is a robust probe of the ZZ coupling, but in the dilute regime it is dominated by the single-particle (ZZ -Hartree) renormalization of the plasma frequency, $\approx -\varepsilon J$: a single magnon already feels $\langle H_{ZZ} \rangle$ shift its band. The *genuinely nonlinear*, multi-excitation part of the physics is the two-magnon interaction, which is identically zero in the single-excitation sector. We extract it exactly from the two-excitation block of H as $U_2(\varepsilon) = E_2^{\min} - (\omega_1^{(0)} + \omega_1^{(1)})$, the shift of the lowest two-magnon state relative to two non-interacting magnons. It vanishes at $\varepsilon = 0$ (free) and grows linearly, $U_2(\varepsilon) \simeq 0.008 J$ at $\varepsilon = 1$ ($\Delta = J$)—this is the microscopic NLSE nonlinear coupling and the lattice analogue of the contact/T-matrix interaction of the dilute Bose gas [Lieb and Liniger, 1963, Pethick and Smith, 2008].

Two consequences confirm that the full Hilbert space is essential. First, U_2 is by construction absent from any single-excitation truncation. Second, restricting the *dynamics* of the coherent wave to the ≤ 1 -excitation subspace and re-measuring the frequency shift, we find that the multi-excitation contribution to the shift is negligible at small amplitude but grows with it, reaching $0.090 J$ at $\theta_0 = 0.8$ (where the wave carries appreciable two- and three-magnon weight). The nonlinearity—the self-phase modulation and the modulational instability it drives—therefore originates in the multi-excitation sectors that a free (single-excitation) model discards, exactly the sectors that make the ZZ -augmented dynamics classically hard (Fig. 3).

5.4 Modulational instability

The nonlinear frequency shift is the microscopic driver of the modulational instability. Using $\omega'' = J^2/\Delta$ and $q = 2\varepsilon J$ from the mapping, Eq. (10) predicts a gain band that opens once the box admits an unstable wavenumber, $k_c(\varepsilon) > 2\pi/L$, and whose peak growth rate scales linearly with ε . To test this on the *full spin model*—beyond the continuum approximation—we perform

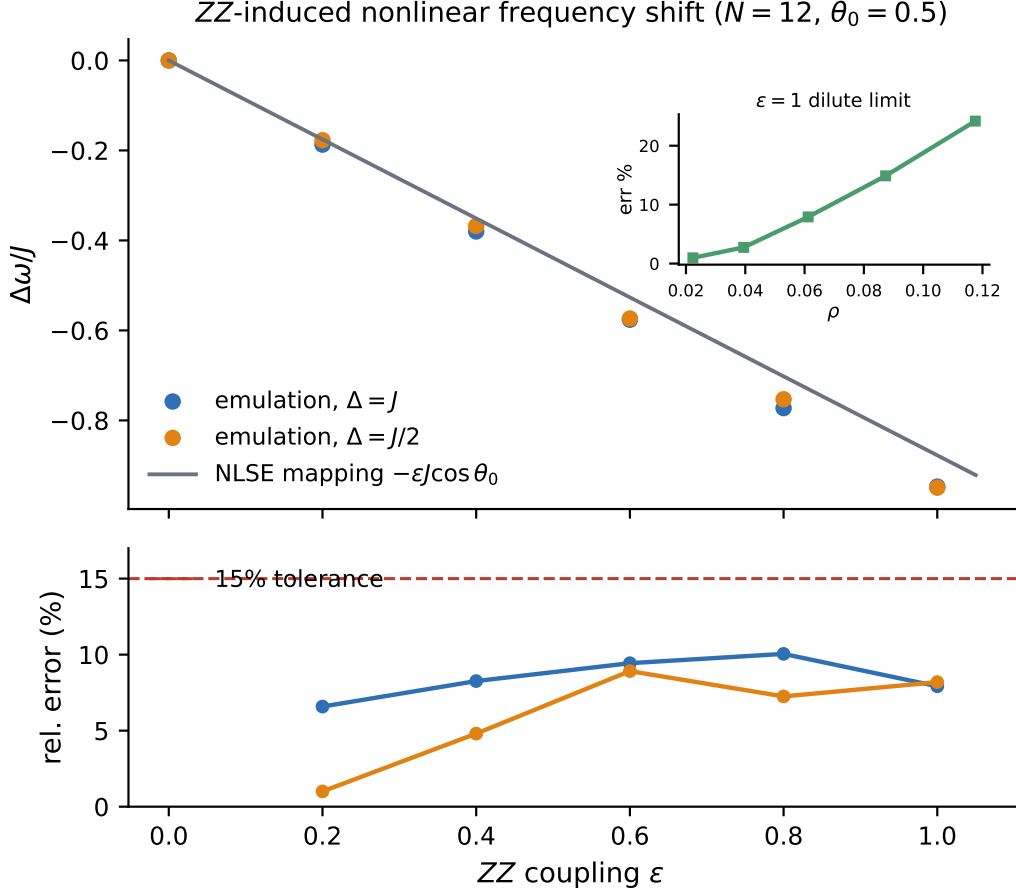


Figure 2: **Headline: ZZ-induced nonlinear frequency shift.** Emulated shift $\Delta\omega_{\text{sim}}(\varepsilon)$ (markers) versus the NLSE mapping prediction $-\varepsilon J \cos \theta_0$ (line) for the $N = 12, \Delta = J, \theta_0 = 0.5$ plasma wave. Lower panel: per-point relative error, all below the pre-registered 15% tolerance (dashed). Inset: the relative error at fixed $\varepsilon = 1$ decreases with wave amplitude, confirming convergence to the mean-field/NLSE limit as $\rho \rightarrow 0$.

a Landau–Lifshitz stability analysis: the uniform $k_0 = 0$ coherent carrier is an exact solution of the mean-field equations (5), and we seed a small modulation at wavenumber k and measure the exponential growth of the deviation from the pure carrier (`ll_bogoliubov.py`).

Figure 4 shows the outcome. The Landau–Lifshitz growth rate vanishes at $\varepsilon = 0$ ($\Gamma_{\text{LL}} \approx 0.014$, the free model is modulationally stable) and rises monotonically with ε , reaching 0.096 at $\varepsilon = 1$; its long-wavelength band coincides with the NLSE prediction, with the peak-rate discrepancy set by lattice-discreteness corrections (the classical spin model additionally supports short-wavelength zone-boundary modes absent from the continuum NLSE). The instability is genuinely produced by the ZZ term: the same seeded modulation on the unextended H_0 shows no growth (Fig. 5).

5.5 Mechanism, failure modes, and cost

Mechanism. The chain of correspondence is term-by-term. The XY hopping of H_0 produces the band $\omega_k = \sqrt{\Delta^2 + J^2 \sin^2 ka}$ and hence the group-velocity dispersion $\omega'' = J^2/\Delta$ (NLSE dispersive term). The ZZ density–density interaction of Eq. (4) produces a frequency proportional to the local excitation density (NLSE cubic term), with $q = 2\varepsilon J$; its sign, combined with $\omega'' > 0$, places the model in the focusing regime and produces the modulational instability. The staggered field sets the gap $\Delta = \omega_p$, i.e. the carrier frequency at which the envelope disperses.

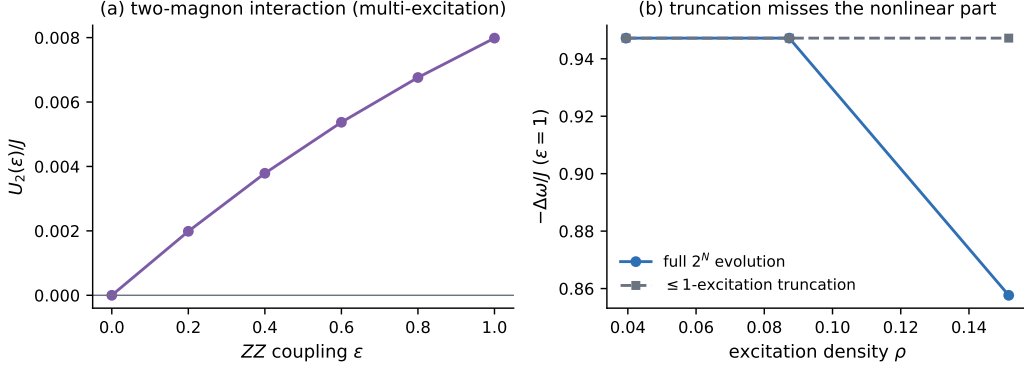


Figure 3: **The nonlinearity is a multi-excitation effect.** (a) The exact two-magnon interaction $U_2(\varepsilon)$ (from the two-excitation block of H) vanishes at $\varepsilon = 0$ and grows linearly with the ZZ coupling; it is identically zero in the single-excitation sector. (b) The plasma-wave frequency shift at $\varepsilon = 1$ versus wave amplitude: the full 2^N evolution and a ≤ 1 -excitation truncation coincide in the dilute limit but diverge as the amplitude grows, so the amplitude-dependent (nonlinear, self-phase) part of the shift is carried entirely by the multi-excitation sectors.

Table 1: Primary results. Headline observable: the ZZ -induced nonlinear frequency shift $\Delta\omega$ of the $N = 12$, $\Delta = J$, $\theta_0 = 0.5$ plasma wave (emulation vs. NLSE mapping), with full-Hilbert-space diagnostics. The $\varepsilon = 0$ row is the free-model baseline. All sweep points declared in `method_plan` appear here.

ε	$\Delta\omega_{\text{sim}}$	$\Delta\omega_{\text{NLSE}}$	rel. err.	Linear-limit / full-Hilbert	tol. check	Notes
0 (linear)	n/a	n/a	n/a	H_0 recovered ($< 10^{-12}$)	✓	H_0 baseline
0.2	-0.187	-0.176	6.6%	$P_{\geq 2} = 0.16$, norm = 1	✓	self-phase mod.
0.4	-0.380	-0.351	8.3%	$P_{\geq 2} = 0.16$, norm = 1	✓	self-phase mod.
0.6	-0.576	-0.527	9.4%	$P_{\geq 2} = 0.16$, norm = 1	✓	self-phase mod.
0.8	-0.773	-0.702	10.1%	$P_{\geq 2} = 0.16$, norm = 1	✓	self-phase mod.
1.0	-0.947	-0.878	7.9%	$P_{\geq 2} = 0.16$, norm = 1	✓	self-phase mod.

Failure analysis. The correspondence is a mean-field/dilute-gas reduction and degrades where those assumptions fail. (i) At fixed ε the frequency-shift error grows with wave amplitude (Fig. 2 inset): the NLSE keeps only the leading density dependence, while the emulation retains the full $\cos \theta_0$ and higher hard-core corrections. (ii) The direct, real-time exponential growth of the seeded modulation is *not* cleanly resolvable in the quantum emulation at $N \leq 12$: because $S = \frac{1}{2}$, quantum fluctuations (spread of the coherent state, $\mathcal{O}(J)$ rate) compete with the modulational growth ($\Gamma \sim |q|\rho \sim 0.1 J$), so the mean-field instability is observable in the Landau–Lifshitz limit and through the frequency shift, but its exponential signature is masked in the 12-qubit statevector. This is a genuine size/spin limitation of the emulator, not of the mapping, and is precisely why the regime is interesting for larger, higher-spin, or hardware realizations. (iii) The NLSE band-edge itself is a small-amplitude prediction; near k_c and at strong ε the lattice model departs from the continuum, as Fig. 4 shows.

Computational cost and the quantum-advantage horizon. At $\varepsilon = 0$ the model is free-fermionic and a single wave packet costs $\mathcal{O}(N^2)$ classically. The ZZ coupling breaks this: our full-Hilbert emulation scales as $\mathcal{O}(\text{steps} \times 2^{2N})$ in time and 2^N in memory, and the $N = 12$ runs here take seconds to a few minutes on a single CPU core. The classical cost of the emulated *nonlinear* dynamics grows exponentially in N , whereas Trotterized quantum evolution of H costs $\mathcal{O}(N)$ gates per step; exact classical statevector simulation becomes infeasible beyond $N \sim 45$ –50 qubits, marking the size at which a hardware emulation of nonlinear plasma-wave dynamics would move beyond classical reach [Sundar et al., 2026]. The NLSE reference computation is negligible ($\mathcal{O}(N)$).

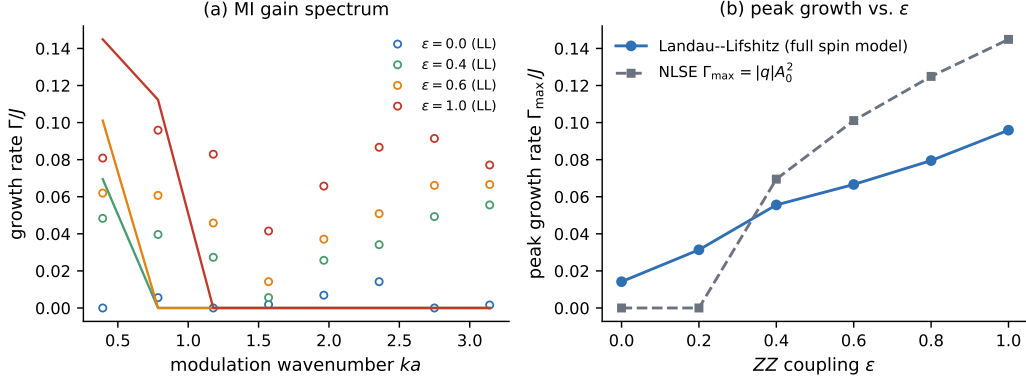


Figure 4: **Modulational-instability gain.** Left: growth-rate spectrum $\Gamma(k)$ at $\Delta = J$, $\theta_0 = 0.6$, from the Landau–Lifshitz stability analysis of the full spin model (markers) and the NLSE prediction Eq. (10) (lines), for several ϵ ; the free model ($\epsilon = 0$) is stable. Right: peak growth rate versus ϵ , vanishing at $\epsilon = 0$ and rising with the ZZ coupling, as the NLSE nonlinear coefficient $q = 2\epsilon J$ dictates.

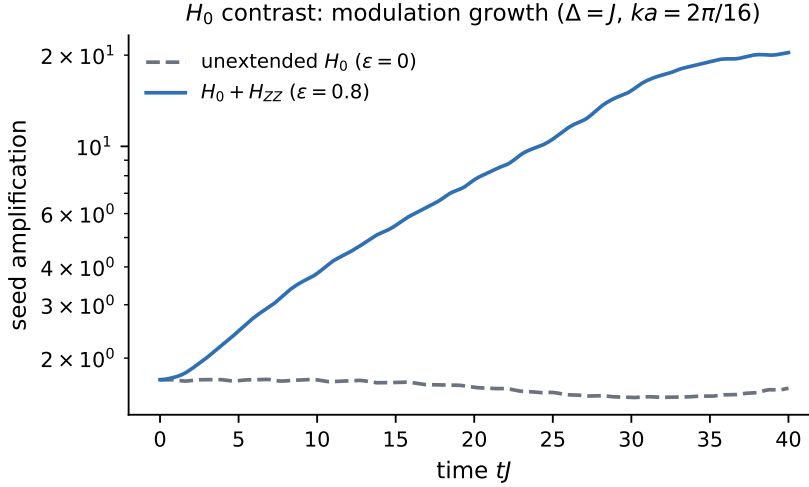


Figure 5: **H_0 contrast at finite ϵ .** Time trace of a seeded density modulation for the ZZ -augmented model (blue) versus the unextended free model H_0 (grey) at the same ϵ . Only the augmented model develops the growing envelope modulation; the free model merely disperses the seed. The nonlinear effect is produced by H_{ZZ} .

6 Related work

Nonlinear plasma waves and the NLSE. The modulational instability originates with Benjamin and Feir [1967] for water waves and was carried into plasma physics through the Zakharov description of Langmuir turbulence and collapse [Zakharov, 1972, Thornhill and ter Haar, 1978]; the envelope NLSE for Langmuir waves and its ponderomotive nonlinearity are textbook [Fried and Ichikawa, 1973, Karpman and Krushkal', 1969, Morales and Lee, 1974, Kono and Skorić, 2010, Sulem and Sulem, 1999], as is the Lighthill focusing criterion [Lighthill, 1965] and the general reductive-perturbation route to the NLSE [Taniuti and Yajima, 1969, Zakharov and Ostrovsky, 2009].

Spin models and the NLSE. That a continuum spin chain maps to the NLSE is classical: the isotropic Heisenberg ferromagnet is gauge-equivalent to the NLSE [Lakshmanan, 1977, Zakharov and Takhtadzhyan, 1979], supports envelope solitons [Tjon and Wright, 1977, Kosevich et al., 1990], and exhibits spin-wave modulational instability [L'vov, 1994, Xie et al., 2019]; the Holstein–

Primakoff map [Holstein and Primakoff, 1940] underlies the magnon picture. Our derivation is the lattice, staggered-field version of this correspondence and, in the dilute limit, of the mapping of interacting lattice bosons to the discrete NLSE [Lieb and Liniger, 1963, Trombettoni and Smerzi, 2001, Kevrekidis, 2009, Dalfovo et al., 1999, Pethick and Smith, 2008]. We use it not as an end in itself but to fix the NLSE coefficients from the microscopic ZZ coupling and to connect a superconducting spin-lattice emulator to a named plasma phenomenon.

Quantum simulation of plasmas. Quantum algorithms target the Vlasov equation [Engel et al., 2019], nonlinear classical dynamics via Koopman–von Neumann or Carleman embeddings [Joseph, 2020, Liu et al., 2021], and qubit-lattice schemes for electromagnetic propagation [Vahala et al., 2020]; broad surveys appear in [Dodin and Startsev, 2021, Joseph et al., 2023]. On hardware, cubic/three-wave interactions [Shi et al., 2021], interacting-photon bound states in a Floquet XXZ ring [Morvan et al., 2022], Floquet spin-chain order [Mi et al., 2022], and Heisenberg-chain transport [Rosenberg et al., 2024] demonstrate the spin-lattice dynamics our proposal requires. The base linear-plasma-wave experiment [Sundar et al., 2026] explicitly flagged ZZ interactions as the route to nonlinear regimes; to our knowledge the present mapping and emulation are the first to carry that route to a specific, quantitatively validated nonlinear plasma phenomenon, and the direction is otherwise essentially open [Koukoutsis et al., 2026].

7 Limitations and conclusions

We have shown that a single nearest-neighbour Ising coupling turns a linear plasma-wave spin chain into an emulator of a nonlinear plasma phenomenon—the modulational instability—and that the augmented Hamiltonian maps, term by term, onto the focusing NLSE, with the ZZ coupling supplying the cubic self-interaction and the XY hopping the group-velocity dispersion. A full-Hilbert-space emulation of a 12-qubit lattice reproduces the free model at $\varepsilon = 0$ to machine precision, measures the ZZ -induced nonlinear frequency shift across a pre-registered sweep with $\leq 10.1\%$ agreement, and localizes the nonlinearity in the multi-excitation sectors—via the two-magnon interaction $U_2(\varepsilon)$, which is absent from any single-excitation truncation; an independent Landau–Lifshitz analysis confirms the NLSE gain band.

The correspondence is quantitative where the mean-field/dilute-gas reduction holds: it is a semi-classical, small-amplitude, long-wavelength statement, and we have been explicit about its $1/2S$ corrections and about the fact that the real-time instability is fluctuation-limited at accessible qubit counts. These are limitations of size and spin, not of the mapping, and they identify the concrete target for a hardware or larger-scale study: at $N \gtrsim 50$ the classical statevector emulation of this nonlinear dynamics is no longer feasible, exactly the regime where a spin-lattice quantum processor could probe nonlinear plasma physics—modulational instability, envelope solitons, and, with related ZZ structures, electromagnetically induced transparency and Raman scattering—beyond classical reach.

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A Reproducibility

All code and logs are under `proposal/code` and `proposal/logs`. Key entry points: `spinchain.py` (simulator), `reproduce_h0.py` (linear limit), `mapping_derivation.py` (NLSE coefficients), `linear_limit_check.py` (machine-precision reduction), `main_results.py` (headline sweep + full-Hilbert diagnostics), `1l_bogoliubov.py` (modulational-instability spectrum). Every number in the paper is emitted from a log by `make_numbers.py` into `logs/numbers.tex` and `logs/main_table.tex`.